BHARATH KUMAR J

DevOps & Cloud Infrastructure Engineer

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PROFESSIONAL SUMMARY

DevOps & Cloud Infrastructure Engineer with **2+ years of enterprise experience at Hewlett Packard Enterprise (HPE)**, specializing in Kubernetes orchestration, CI/CD pipeline automation, Infrastructure as Code (Terraform), and cloud observability. Proven track record administering 100+ production server nodes maintaining **99.9% uptime SLA**, engineering internal FastAPI automation services to **cut operational overhead by 35%**, and driving down incident **MTTR by 25%**. Developer of autonomous AI-powered SRE monitoring systems leveraging the Model Context Protocol (MCP).

TECHNICAL SKILLS

Cloud & Platforms:	AWS (EC2, S3, IAM, VPC), Google Cloud Platform (GCP), HPE GreenLake, Enterprise Linux (RHEL/Ubuntu), Networking & DNS
Containers & K8s:	Kubernetes (K8s), Docker, Helm, Horizontal Pod Autoscaler (HPA), Containerd, Kind, Minikube
CI/CD & IaC:	GitHub Actions, Terraform, Jenkins, Ansible, YAML Declarative Pipelines, GitOps, Git
Observability & SRE:	Prometheus, Grafana, ELK Stack / Loki, Incident Triage, Root Cause Analysis (RCA), MTTR Optimization, MCP
Automation & Security:	Python 3, FastAPI, Bash Scripting, REST APIs, Redis, Pytest, DevSecOps (Trivy), Cloud IAM & RBAC Governance

WORK EXPERIENCE

Hewlett Packard Enterprise (HPE)

Bengaluru, India

DevOps & Automation Specialist *(Official HR Title: Technical Solutions Specialist)* — **Jan. 2025 – Present**

- Administered and sustained 100+ bare-metal and hybrid-cloud enterprise server nodes, maintaining **99.9% production availability** across high-availability clusters.
- Engineered automated Python scripts and high-concurrency **FastAPI REST APIs** for self-service node auditing, eliminating **35% of repetitive manual sysadmin workflows**.
- Championed centralized observability and log analytics (ELK/syslog), conducting deep Root Cause Analysis (RCA) to drive down Mean Time to Resolution (**MTTR) by 25%**.
- Collaborated with software engineering squads to govern zero-downtime releases, standardize configuration templates, and validate post-rollout health metrics.

Cloud Infrastructure & Automation Engineer — **Feb. 2024 – Dec. 2024**

- Governed enterprise cloud Identity & Access Management (IAM) across multi-tenant environments, enforcing least-privilege RBAC controls for **500+ enterprise accounts**.
- Built automated Bash and Python cron workflows for scheduled audit compliance, log rotation, and credential expiration alerts—saving **15+ engineering hours weekly**.
- Investigated authentication anomalies, permission drifts, and system bottlenecks through systematic log aggregation and telemetry diagnostics.
- Authored operational runbooks and infrastructure configuration templates to ensure high audit compliance across hybrid cloud infrastructure.

KEY PROJECTS

GCP Sentinel — Autonomous AI Infrastructure Monitoring Agent

github.com/Bharath-kumar-J/gcp-sentinel

Tech Stack: Python 3.12, Model Context Protocol (MCP), DeepSeek AI, GCP APIs, SQLite, Pytest (99 Tests)

- Built an autonomous AI monitoring agent watching Google Cloud infrastructure through an MCP server equipped with **15 custom read-only inspection tools**.
- Implemented deterministic detection rules for CPU/Memory spikes, Pod CrashLoopBackOff, container OOM kills, Cloud SQL saturation, and load balancer 5xx errors.
- Integrated DeepSeek AI reasoning for automated root-cause analysis (RCA) with automatic fallback to rule-based heuristic reasoning when offline.
- Developed alert de-duplication with SQLite-backed fingerprinting, configurable cooldown periods, and **99 automated tests** covering detection loops.

Cloud-Native GitOps & CI/CD Platform

github.com/Bharath-kumar-J/capstone-devops

Tech Stack: GitHub Actions, Docker, Kubernetes (Kind), Terraform, Prometheus, Grafana, Flask, Redis, Trivy

- Architected an automated CI/CD pipeline using GitHub Actions, featuring automated linting, Pytest validation, and Trivy container vulnerability scanning.
- Containerized microservices using multi-stage Docker builds and deployed onto Kubernetes with Horizontal Pod Autoscaler (HPA) dynamically scaling replicas.
- Provisioned cluster networking, storage classes, and compute instances declaratively using reusable Terraform modules.
- Integrated Prometheus metric scrapers and configured custom Grafana dashboards tracking real-time cluster telemetry, error budget, and service latency.

EDUCATION

Sathyabama Institute of Science and Technology — Chennai, India

Aug. 2019 – May. 2023

Bachelor of Engineering (B.E.) — Electronics and Communication Engineering